THE NATIONAL UNIVERSITY OF LESOTHO

Interface Lead: Interface Design and Testing Guide

Role document for Milestone M2 · Qaba Moiketsi

Team	limo-nul (lab group C)	**Release tag**	m2
Role	**Interface lead**	**Member**	Qaba Moiketsi (202002573)
GitHub	qabamothe	**Due**	Sun 11 Oct 2026, 23:59

1 Your role in M2

- Deliver the **interface** where programs are typed, assembled and checked: `index.html`, `src/ui.js`, `src/style.css` (HTML/CSS/JavaScript: required by the handout; no Java is needed).
- Deliver the **wireframes** of all views, including the M3/M4 views.
- Keep the site small (now about 55 KB; limit 2 MB), offline-capable, and usable without relying on colour.

2 Where you test your programs: step by step

A. Open the interface

1. **Online:** open https://recocipher-beep.github.io/limo-nul/ in Chrome, Edge or Firefox (after the team enabled GitHub Pages).
2. **Offline, on your own computer:** download the repository as a ZIP, unzip it, then **double-click index.html**. No installation, no server, no internet needed.

B. Test Program 1 (the sum from 1 to 10)

1. Left panel, drop-down **"Load an example..."** → choose **1. Sum from 1 to 10 (loop)**. The program appears in the editor with line numbers (it also loads automatically when the page opens).
2. Press **Assemble** (or Ctrl + Enter). The green bar must say **"Assembled: 7 instructions, 28 bytes."**
3. Right panel, tab **Machine code**: each instruction has its address, hex word, the 32 bits split into *named, coloured fields* and its RV32I equivalent. Check against the table below.
4. Tab **Labels**: LOOP = 12 (0xC), DONE = 28 (0x1C).
5. Tab **Export**: **Copy hex** gives one word per line; **Copy RV32I** gives the program for the Ripes cross-check.
6. **Save .limo file** downloads `program.limo`; **Open .limo file** loads it again.

Address	Instruction	Hex shown
0	`kpnh R5, R0, 1`	`0x00100293`
4	`kpnh R6, R0, 0`	`0x00000313`
8	`kpnh R7, R0, 11`	`0x00B00393`
12	`kpn R6, R6, R5`	`0x00530333`
16	`kpnh R5, R5, 1`	`0x00128293`
20	`khk R5, R7, DONE`	`0x0072D463`
24	`tl LOOP`	`0xFF5FF06F`

C. Test the error messages (type these, press Assemble)

Type in the editor	You must see
`kpm R1, R2, R3`	Line 1: Unknown instruction "kpm". Did you mean "kpn"?
`kpn R1, R2`	Line 1: "kpn" needs 3 operands ... but you wrote 2.
`kpnh R1, R2, 5000`	Line 1: 5000 does not fit in the 12-bit immediate ... -2048 to 2047
`khl R1, R2, NOWHERE`	Line 1: Label "NOWHERE" is not defined ...
(any error)	Red bar "x N errors"; the error line number in the margin has an "x"; clicking a message selects that line

D. Test all 8 programs

Choose each example in turn, press Assemble, and confirm the green bar. The expected instruction counts are in the test report (Program 1: 7 lines of code, 42 executed). The automatic checks are run by the test lead with `node tests/run_tests.js`.

3 Interface design

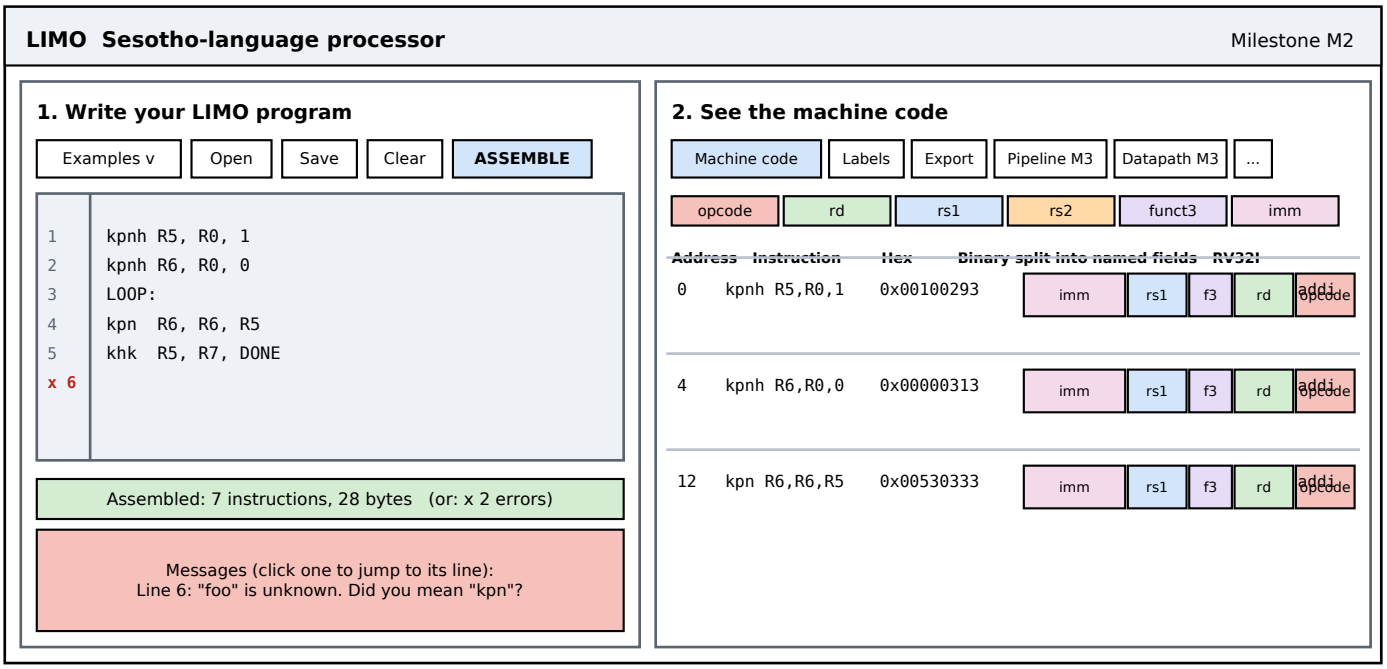


Figure 1: Editor and assembler view (built). Left: write; right: read the machine code.

Element	ID in index.html	What it does
Examples menu	examples	loads one of the 8 programs from src/examples.js
Open / Save / Clear	btn-open, btn-save, btn-clear	file in and out (.limo text), clear editor
Assemble	btn-assemble	calls `LIMO.assembler.assemble(text)`; also runs 0.35 s after typing stops
Editor + margin	editor, gutter	line numbers synchronised with scrolling; error lines marked "x"
Status bar / messages	status, messages	success or error count; one clickable line per error
Machine code table	code-table	address, source, hex, binary fields (named and coloured), RV32I text
Labels, Export	label-table, out-hex, out-rv	label addresses; copy-ready hex and RV32I
Dashed tabs M3/M4	view-pipe, view-data, view-ctrl, view-haz, view-stat	wireframes now, real views later

Wireframes of the M3/M4 views

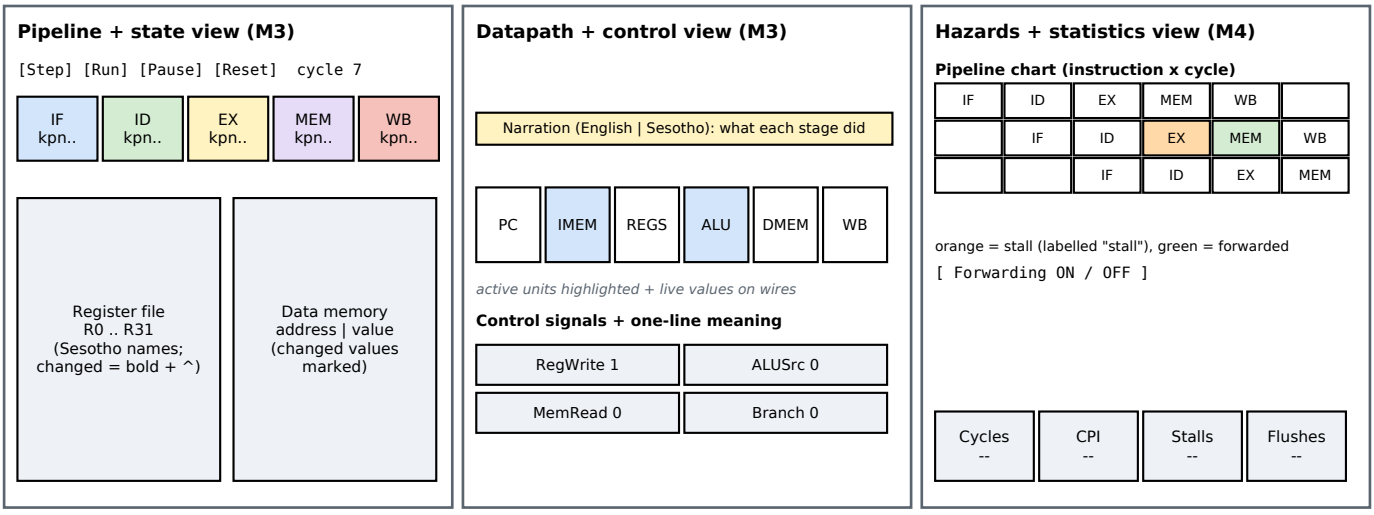


Figure 2: Pipeline + state (M3), datapath + control (M3), hazards + statistics (M4).

4 Accessibility and quality checklist

Requirement (handout B7 / good practice)	How it is met	Check
Colour is never the only cue	every field shows its name as text; errors have a text label, an "x" in the margin and a status word; stall/forward will carry text labels	[ ]
Keyboard use	all buttons focusable with visible outline; Ctrl + Enter assembles; tabs move with the arrow keys	[ ]
Screen readers	status bar is a live region; tabs use role="tab"/"tabpanel"; editor has a label	[ ]
Light and dark mode	colours are CSS variables, dark values under prefers-color-scheme	[ ]
Small screens	two columns become one below 860 px width; tables scroll sideways	[ ]
Size and offline	no libraries, no network requests; about 55 KB in total	[ ]
Slow link / old laptop	plain JavaScript, no frameworks	[ ]

Tick the last column yourself after checking each item on a phone and a laptop.

5 Defence preparation

Likely defence question	Answer you should be able to give
How does the interface talk to the assembler?	`LIMO.assembler.assemble(text)` returns {ok, errors, items, words, labels, hexText, rvSource}; ui.js only displays it.
How is colour not the only cue?	Each field carries its name; errors have text, an "x" and a status word.
Why no framework?	The site must be under 2 MB and load on a slow link; plain JavaScript is about 55 KB.
What will the Datapath view show?	the active units and wires with live values from the pipeline registers of the data model.